

XF-SISS: XOR-Based Fault Tolerant and Flexible Secret Image Sharing Scheme

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Abstract—Sensitive visual data can be protected by splitting it into encrypted fragments recoverable only through collaboration. Traditional (n, n) threshold schemes require all shares for reconstruction, making them impractical in lossy or unpredictable environments. We propose a flexible XOR-based Secret Image Sharing method using Boolean operations to enable incremental, part-wise recovery. This scheme overcomes all-or-nothing limitations while maintaining perfect secrecy. Experiments confirm its fault tolerance and suitability where participant availability is uncertain.

Index Terms—Secret Image Sharing, Flexible SIS, XOR-based, Fault-Tolerant SIS

I. INTRODUCTION

Secret Image Sharing (SIS) is a cryptographic method that safeguards sensitive images by splitting them into multiple fragments known as shares. These shares, when used together, enable the reconstruction of the original image—but only when all of them are present. The (n, n) threshold model is particularly popular due to its simple structure and robust confidentiality, as it ensures that without access to all n shares, no information about the original image can be deduced.

Despite its strong security, this rigid requirement introduces practical limitations. In real-world applications such as secure data archiving, distributed storage, or collaborative systems, ensuring that all participants or data holders are available simultaneously can be unrealistic. If even one share is lost or inaccessible, the image becomes unrecoverable, thereby compromising usability in dynamic environments.

To address these limitations, progressive SIS techniques have been introduced. These schemes allow for partial or degraded reconstruction of the original image using fewer than n shares. However, this added flexibility typically comes at the cost of reduced image quality, increased risk of information leakage, or a lack of precise control over which image regions are reconstructable.

In light of these challenges, we introduce a flexible XOR-based (n, n) SIS scheme that enables targeted, part-wise image reconstruction using selected shares and corresponding private keys. Our method maintains the perfect secrecy guarantees of traditional SIS approaches while supporting exact, lossless recovery of specific image segments—without increasing the

overall image size. This makes it especially suitable for real-world applications where user availability or network reliability is uncertain.

II. RELATED WORKS

Over the years, SIS techniques—particularly those leveraging XOR operations—have evolved significantly. These methods rely on bitwise computations to generate secure shares and efficiently restore the secret image. Most of these techniques conform to the (n, n) threshold structure.

Wang et al. [1] were among the first to use basic logical XOR operations for secure SIS. Subsequently, Chen et al. [2] extended the model to support multiple secrets in an (n, n) context. However, a later variation by Chen and Wu [3] was found by Yang et al. [4] to violate threshold properties. Yang et al. subsequently proposed a corrected and secure revision.

Chen et al. continued to enhance their framework in [5], and Deshmukh et al. [6] proposed a hybrid scheme combining XOR operations with the Chinese Remainder Theorem—improving security at the expense of computational efficiency. In a follow-up work [7], they offered three distinct SIS variants, although one faced criticism from Prasetyo et al. [8] for failing under specific recovery conditions, prompting further refinements.

Other researchers have contributed various enhancements focusing on verifiability, minimal pixel expansion, and randomness improvement in shares [9], [10], [11], [12], [13], [14], [15], [16], [17].

Despite these advancements, a key limitation remains: traditional XOR-based (n, n) SIS schemes lack fault tolerance. If even one share is compromised or missing, the secret becomes entirely unrecoverable. This inflexibility poses a significant challenge in practical scenarios.

Progressive SIS schemes are cryptographic frameworks that support stepwise reconstruction of the hidden image, with image quality improving as more shares are gathered. In contrast to traditional secret image sharing schemes, which only allow full recovery of the image when all shares are present, they introduce a progressive recovery process where even partial availability of shares reveals the image, albeit at

a reduced quality. As more shares are collected, the quality of the reconstructed image progressively improves.

Key Features of Progressive SIS Schemes:

- 1) **Progressive Reconstruction:** The primary innovation is their ability to allow partial recovery of the secret image before all shares are available. Initially, using only a few shares may yield a vague or low-resolution version of the image, with reconstruction quality gradually improving as additional shares are incorporated.
- 2) **Security:** Even though the image becomes partially visible with fewer shares, the security of the scheme is designed such that unauthorized parties still cannot infer significant information unless they reach a predefined threshold. The partial images generated from fewer shares are intentionally degraded in quality to minimize the leakage of sensitive information.
- 3) **Threshold Flexibility:** Progressive schemes generally allow flexibility in setting thresholds for image reconstruction.

They are particularly useful in scenarios like secure data sharing, distributed storage, or disaster recovery, where it is not always possible to gather all shares at once. Instead, a rough or degraded version of the image can be reconstructed progressively, allowing for partial data access while full access remains conditional on more shares being collected. A detailed comparison between **progressive SIS schemes** and **the proposed scheme** is provided in section VI.

These XOR-based SIS schemes have laid the foundation for secure image sharing but fall short in scenarios requiring flexibility. Specifically, they do not support partial recovery of the secret image when only a subset of the shares is available. This limitation is particularly problematic in dynamic environments where participant availability cannot always be ensured. The proposed scheme in this paper addresses these gaps by introducing a flexible (n, n) threshold SIS approach that maintains the perfect security of the traditional schemes while allowing for partial image recovery with fewer than n shares. This flexibility is achieved through a novel share-generation process that utilizes Boolean operations to progressively combine available shares, enabling partial recovery without compromising the overall security.

III. PROPOSED SCHEME

We present a flexible XOR-based SIS method tailored to the (n, n) threshold framework. In this model, n share images are generated from the input image. Section III-A details the share generation phase, while Section III-B explains the reconstruction process. The design emphasizes part-wise recoverability and robust secrecy, even when only a subset of shares and their private keys are accessible.

A. Sharing Phase

Input to this phase is a Secret Image, A , of size $w \times h$. The output of the phase is a public image PK of size $w \times h$, n private images $K_i, i = 1, \dots, n$, each of size $w \times h$, and n share images S_i each of size $w \times h$ for $i = 1 \dots, n$. The

procedure for sharing is as below.

Sharing Procedure:

- 1) The secret image A needs to be partitioned into n partitions, I_i , either horizontally or vertically as follows:

Horizontal Partitioning:

$$\text{Let } t = \lfloor \frac{w}{n} \rfloor \quad i = 1, 2, \dots, n$$

$$I_i(x, y) = \begin{cases} A(x, y) & \text{if } (i-1) \times t \leq x < \min(i \times t, w) \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Vertical Partitioning:

$$\text{Let } t = \lfloor \frac{h}{n} \rfloor \quad i = 1, 2, \dots, n$$

$$I_i(x, y) = \begin{cases} A(x, y) & \text{if } (i-1) \times t \leq y < \min(i \times t, h) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

While this work primarily uses uniform horizontal or vertical partitioning, the scheme can be trivially extended to support non-uniform partitions.

- 2) After partitioning the image, n purely random grayscale images, K_i of size $w \times h$, where $i = 1, 2, \dots, n$, are generated which are used as the private images for respective shares.
- 3) Next the noised images, N_i , are generated as follows:

$$N_i = F_2(F_1(I_i \oplus K_i)) \quad \text{where } i = 1, 2, \dots, n \quad (3)$$

where F_1 is defined as the *circular_bit_shift* function and F_2 is the *reverse_bit* function.

$$F_1(I) = \text{circular_bit_shift}(I(x, y), (x + y) \bmod 8)$$

$$F_2(I) = \text{reverse_bit}(I(x, y)) \quad (4)$$

The functions F_1 and F_2 serve to enhance randomness and obscure any residual structure in the intermediate image.

- 4) The share images, S_i , can be obtained as follows:

$$S_i = N_i \oplus PK \quad \text{where } i = 1, 2, \dots, n \quad (5)$$

where PK is the randomly generated public image which is generated at the time of sharing and is

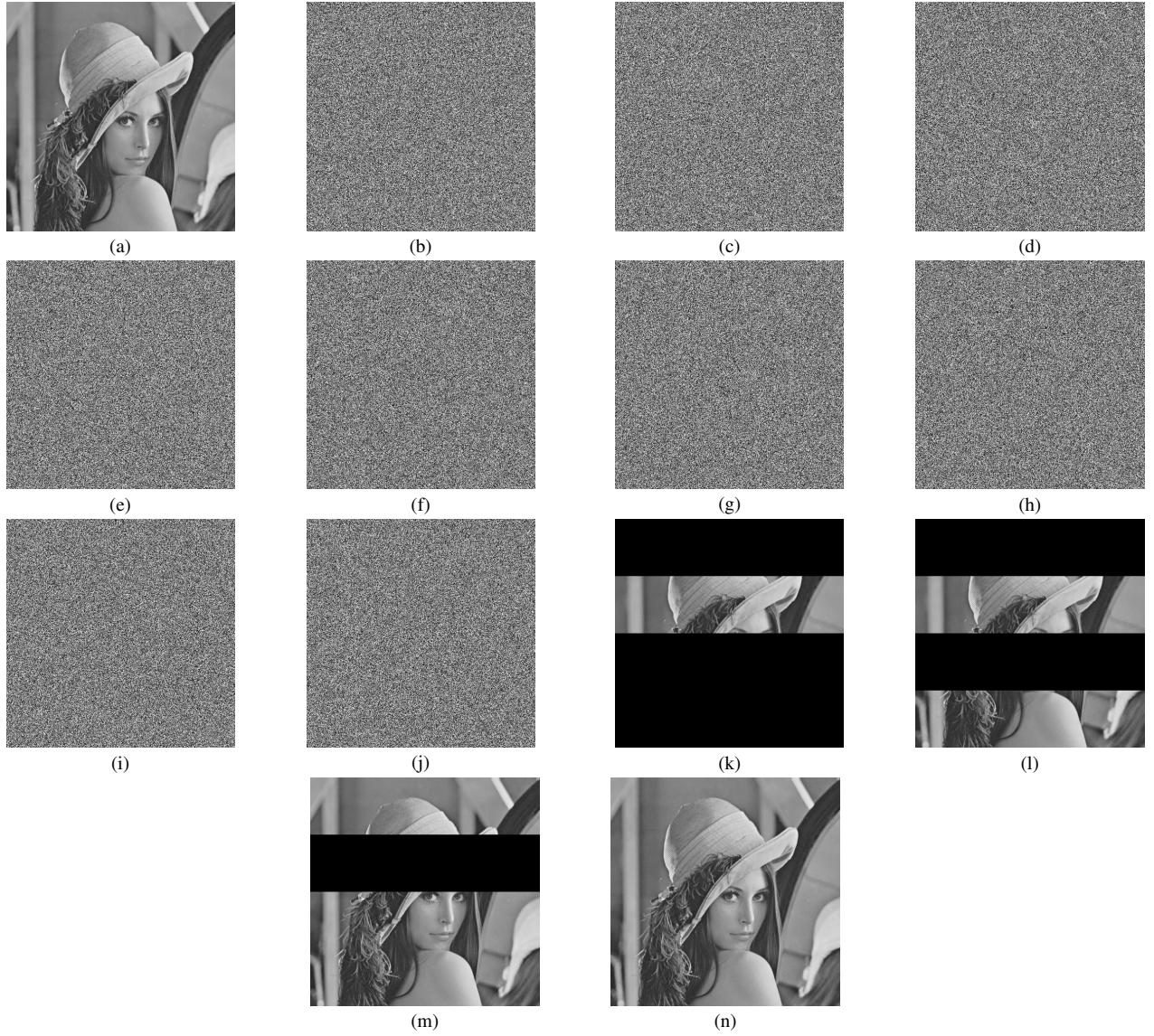


Fig. 1. (a) Standard Lena Image, (b)-(e) Generated share images, (f) Generated public image, (g)-(j) Generated private images, and (k)-(n) Recovered images using any 1, 2, 3, and 4 shares, respectively.

accessible to the public.

- 5) Each participant, P_i , receives a tuple $\{S_i, K_i\}$, where $i = 1, 2, \dots, n$ - The set of a share image and the corresponding private image.

B. Recovery Phase

Input to this phase is a public image PK , any m share images S_i , and corresponding m private images K_i each of size $w \times h$. The output of the phase is a partially/completely reconstructed image.

Recovery Procedure:

- 1) The participants that participate in the recovery phase bring their respective share images, S_i , and private images, K_i , where $i \in \{1, 2, \dots, m\}$

- 2) The noised images, N_i , can be obtained as follows:

$$N_i = S_i \oplus PK \quad \text{where } i \in \{1, 2, \dots, m\} \quad (6)$$

The public image, PK , is available on the cloud.

- 3) Next all the noised images obtained from (6) are aggregated to obtain a single noised image.

$$N = N_1 \oplus N_2 \oplus \dots \oplus N_m \quad (7)$$

- 4) The recovered image, $\bar{A}$, can be obtained as follows:

$$\bar{A} = F_1(F_2(N)) \oplus K \quad \text{where } i \in \{1, 2, \dots, m\} \quad (8)$$

TABLE I

(A) FOR THE (4, 4) LENA EXPERIMENT,

Number of involved participants	PSNR against reconstructed image	PSNR against recovered part
1	7.0355	∞
2	8.7694	∞
3	11.2821	∞
4	∞	∞

(a)

(B) FOR THE (6, 6) BABOON EXPERIMENT

Number of involved participants	PSNR against whole reconstructed image	PSNR against recovered part
1	6.5403	∞
2	7.6751	∞
3	8.5629	∞
4	10.8311	∞
5	13.7369	∞
6	∞	∞

(b)

where F_1 and F_2 are the same functions defined in (4) and K is obtained by aggregating all the available private keys.

$$K = K_1 \oplus K_2 \oplus \dots \oplus K_m \quad (9)$$

5) $\bar{A}$ is the recovered image.

IV. EXPERIMENTAL RESULTS

To evaluate our proposed XOR-based flexible SIS technique, we conducted two sets of experiments: one using a (4, 4) configuration and another using a (6, 6) setup. These test cases allowed us to measure recovery accuracy, image fidelity, and operational overhead under varied conditions.

Fig. 1 illustrates the results of the proposed (4, 4) flexible SIS scheme using Lena as the secret image A . The image was first horizontally divided into intermediate partitions as per Equation (1). Four share images, S_1 , S_2 , S_3 , and S_4 , are shown in Fig. 1(b)–(e), respectively. The corresponding public image PK is depicted in Fig. 1(f). Each share has an associated private image: K_1 , K_2 , K_3 , and K_4 , shown in Fig. 1(g)–(j), respectively.

We evaluated different reconstruction scenarios to demonstrate the scheme's flexibility. Fig. 1(k) presents the output when only S_2 is available. Fig. 1(l) shows the result with S_2 and S_4 , and Fig. 1(m) shows the recovery when S_1 , S_3 , and S_4 are available. Full recovery using all four shares is illustrated in Fig. 1(n).

Fig. 2 shows a similar experiment that was performed on the Baboon image using vertical partitioning (as per (2)). The scheme correctly reconstructed corresponding parts of the image for different combinations of available shares, illustrating the scalability of the approach to higher values of n .

A. Evaluation

Table I presents the PSNR values for both experiments. When the PSNR was computed over the entire image, including unrecovered (black) regions, the values were low due to high MSE in those areas. However, when the PSNR was

computed only over the recovered regions, the MSE was effectively zero, leading to theoretically infinite PSNR. This demonstrates that the recovered regions are identical to the original image segments, validating the lossless nature of the reconstruction.

As more shares become available, the proportion of the image that can be accurately recovered increases, improving the overall PSNR. However, the partial PSNR remains infinite for any correctly recovered region, highlighting the precision and deterministic reconstruction property of the proposed SIS scheme.

V. SECURITY ANALYSIS

A. Threat Model

We assume a threat model where an adversary may obtain some share images but not the corresponding key images, as the keys are securely managed using trusted hardware modules like TPMs. In such a case, the adversary can at most compute a noised image, which—by design—leaks nothing about the original image without the matching key.

B. Security Properties

The proposed scheme provides perfect secrecy in the information-theoretic sense. Specifically:

- Each segment I_i is masked using a truly random K_i , producing $I_i \oplus K_i$. Without knowledge of K_i , the masked segment is statistically indistinguishable from random noise.
- The circular bit shift and bit-reversal transformations add nonlinearity and are only invertible with the correct K_i , preventing structure leakage in the absence of the key.
- The public image PK adds <https://www.cybernoweducation.com/soc-analyst-jobsno> advantage to an adversary without the corresponding K_i , since:

$$S_i = N_i \oplus PK = F_2(F_1(I_i \oplus K_i)) \oplus PK$$

The output appears random and cannot be reversed without knowledge of K_i .

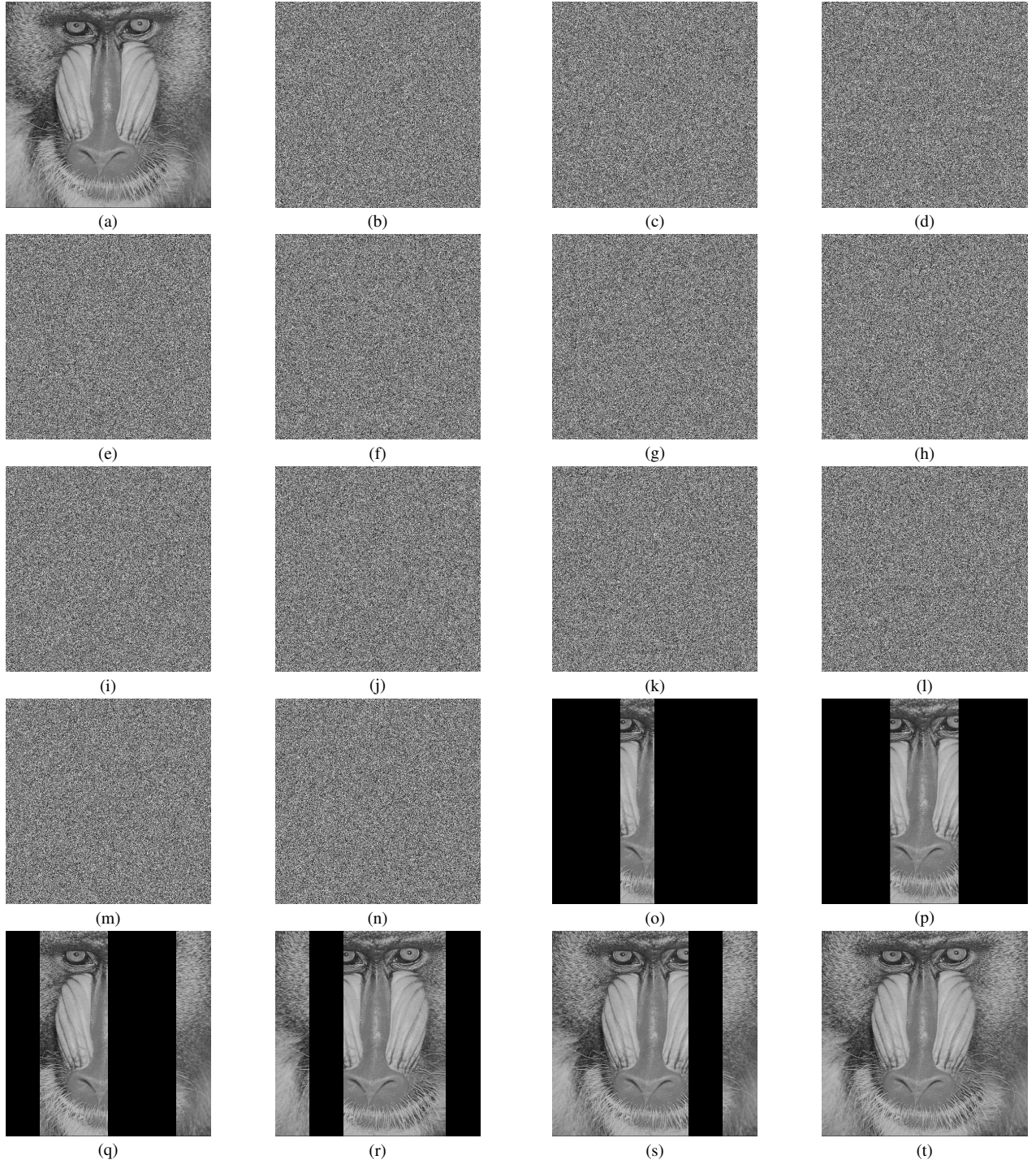


Fig. 2. (a) Standard Baboon Image, (b)-(g) Generated share images, (h) Generated public image, (i)-(n) Generated private images, and (o)-(t) Recovered images using any 1, 2, 3, 4, 5, and 6 shares, respectively.

Therefore, even if some unauthorized shares are accessed and paired with the public image PK, they reveal nothing about the original image without access to the matching private keys.

VI. COMPARISON

The proposed scheme distinguishes itself from traditional XOR-based SIS techniques by addressing their core limitations. This section presents a comparative analysis focusing

on key aspects like image fidelity, share randomness, reconstruction flexibility, and threshold security. A major strength of the proposed method is its ability to ensure lossless recovery—perfectly reconstructing the original image when required shares and private keys are present. It also enhances share randomness, making them harder to reverse-engineer. Crucially, the scheme offers flexible recovery. Unlike con-

TABLE II
COMPARISON BETWEEN THE PROPOSED SCHEME AND OTHER BOOLEAN BASED SIS SCHEMES

Schemes	Number of shared images	Lossless recovery	Pixel expansion	Randomness of shares	Flexible recovery	Threshold security
Wang et al. [1]	n	Yes	No	Yes	No	Yes
Ou et al. [17]	n	Yes	No	Yes	No	Yes
Chattopadhyay et al. [18]	n	Yes	No	Yes	No	Yes
Prasetyo and Hsia [16]	n	Yes	No	Yes	No	Yes
Soreng and Kandar [19]	n	Yes	No	Yes	No	Yes
Wu and Yao [20]	n	Yes	Yes	Yes	No	Yes
The proposed scheme	n	Yes	No	Yes	Yes	Yes

ventional XOR-based (n, n) schemes that demand all shares for recovery, the proposed approach allows exact, part-wise recovery when only a subset of shares and corresponding private keys are available. This flexibility makes it more practical for real-world settings where all participants may not always be accessible. Despite this adaptability, the method retains threshold security: without the complete set of shares and private images, the secret cannot be reconstructed—ensuring strong confidentiality. Compared to progressive SIS schemes, which gradually enhance overall image quality as more shares are collected, the proposed scheme takes a targeted approach. It restores specific parts of the image in full quality as soon as their corresponding share and key are available. Thus, instead of a blurry-to-clear progression, it offers precise and immediate recovery of available segments without approximation.

Table II provides a detailed comparison, illustrating how the proposed method performs across these crucial parameters relative to existing XOR-based SIS techniques.

VII. CONCLUSION

This paper proposed a flexible XOR-based (n, n) SIS scheme supporting both full and part-wise image reconstruction using Boolean operations and randomized transformations. The method ensures perfect secrecy and lossless recovery, even when some shares are missing. Experimental results and comparisons confirm its security, effectiveness, and real-world practicality.

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